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ABSTRACT

INTRODUCTION

The purposes of this study were therefore twofold. Firstly, the study aimed to evaluate the time period for which a standard power training protocol would yield improvements in power output. Secondly, the study sought to

ascertain if training at the optimal load would lead to greater gains in power production than training with a load just below the optimal load.

METHODS

Experimental Approach to the Problem

Twenty-one resistance-trained men were tested for their 1 repetition maximum (1RM) and maximum power output at loads from 30 to 80% of their 1RM (ie, their power profile) in the bench pull (BP). Their peak power output and the load at which it was achieved (ie, the optimal load) were identified from the power profile. Subjects were then randomly assigned to 2 groups and performed 5 sets of 3 repetitions at either their optimal load or at 10% of their 1RM below their optimal load twice per week for 7 weeks. The subjects' power profiles were reassessed after 4 weeks of the training programme and the training loads adjusted to ensure that the subjects were still training at the designated load for the final 3 weeks of the training programme. Finally, the subjects were retested for their 1RM and power profile at the end of the training intervention. The experimental design is depicted in Figure 1.

The BP exercise was chosen for this study as it is easy to perform and removes questions about including bodyweight in the calculations that arise in exercises like loaded jumping (35). It also allows the subject to accelerate the load throughout the whole lift, thereby removing the potential variance associated with the deceleration or projection of the load at the end of the concentric phase (9). Finally, the BP is a commonly used testing and training exercise across a range of different sports (1,15,16,21,27).

Subjects

Twenty-five male university athletes with at least 1 year of resistance training experience were recruited to take part in this study. The inclusion criteria stipulated that all subjects reported no upper-body injuries or pathology in the 12 months before data collection. Subjects were randomly divided into a group that trained at their optimal load (P) and a group that trained with a weight 10% of their 1RM below their optimal load (P-). In addition, subjects had to complete 85% of

the training sessions to be included in the final analysis and 4 subjects were excluded for not meeting this criterion (thus for P, $n = 9$ and for P-, $n = 12$). There were no significant ($p \leq 0.05$) differences in the subject characteristics at the start of the study (Table 1). Ethical clearance was granted by the Institutional Review Board of St Mary's University College, and subjects provided written informed consent.

Sample Size Determination

The appropriate sample size was determined by considering the power of the study to detect differences in the peak power output between the 2 groups across the 3 time points. In particular, we determined the sample size necessary to achieve a power of 0.80 with a significance criterion of $\alpha = 0.05$. To calculate the sample size, we used the equation of Overall and Doyle (31) for longitudinal designs, assuming that the increase in power output would be linear across the course of the study and that there would be no difference between the 2 groups at the start of the study. In addition, we assumed that the correlation of the repeated measures was $\rho = 0.6$ (18). We assumed that the variation (ie, the *SD*) in the peak power output of the participants would be 35 W. This was based on the results of Baker (5) who found that the *SD* of power output scores in the bench throw ranged from 19 to 32 W depending on the ability and age level of the subjects (in particular, the 36 university aged athletes who would be most similar to our cohort had a *SD* of 32 W). We also assumed that the average increase in power output over the course of the study for the cohort as a whole would be 10%, based on the consideration of several short-term upper-body power training studies (19,20,28). Our experience with this cohort of athletes lead us to estimate that the initial mean peak power output would be 650 W; thus, we estimated that the average increase in power output for the cohort as a whole would be 65 W. We therefore defined the minimum expected difference between the 2 groups to be half this value, that is, 32.5 W. These values lead to a required sample size of 14.6 subjects in each group, which we rounded up to 15. We thus aimed to recruit 30 total subjects; however, unfortunately we fell a little short of this number.

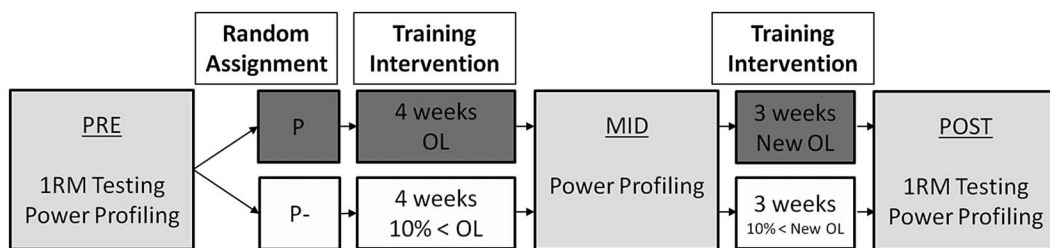


Figure 1. The experimental design used in this study.

TABLE 1. Subject characteristics given as mean $\pm$ SD.*†

	P Group	P- Group
<i>n</i>	9	12
Age (y)	21.7 $\pm$ 2.1	21.8 $\pm$ 1.5
Height (m)	1.84 $\pm$ 0.07	1.84 $\pm$ 0.08
Weight (kg)	83.9 $\pm$ 14.2	88.3 $\pm$ 11.7
1RM (kg)	73.1 $\pm$ 14.2	83.5 $\pm$ 10.0
Peak power output (W)	602 $\pm$ 128	672 $\pm$ 69
Optimal load (%1RM)	64.4 $\pm$ 7.3	64.2 $\pm$ 10.0

*1RM, 1 repetition maximum.

†There were no significant differences between the groups ($p \leq 0.05$).

Instrumentation

A linear force transducer (TENDO Weightlifting Analyzer; TENDO Sports Machines, Trencin, Slovak Republic) was used to record the maximum power output during each repetition. The TENDO unit has been previously assessed and shown to be accurate and reliable in the measurement of power in explosive movements (22,36). The unit was attached to the center of the bar and placed directly beneath it to ensure a vertical cable alignment.

Procedures

Subjects completed initial testing in the first week (PRE; 1RM test followed by a day's rest and then power profiling). They then trained for 4 weeks using either the optimal load identified from the power profile (P) or a load 10% of their 1RM below their optimal load (P-). After this, the subjects'

TABLE 3. An example of a typical programme completed by the subjects.

Exercise	Sets $\times$ reps	Rest (s)
Session 1		
Stretch, warm-up*		
1. Bench Hip Lift	3 $\times$ 8	30
2a. Hang Clean	4 $\times$ 4	60
2b. Box Jump	4 $\times$ 3	90
3. Front Squat	1 $\times$ 10, 1 $\times$ 8, 1 $\times$ 6	120
4. Bench Pull	5 $\times$ 3	180
5a. Bench Press	3 $\times$ 8	30
5b. Plank March	3 $\times$ 8	30
5c. Side Plank Plus	3 $\times$ 8	30
Stretch		
Session 2		
Stretch, warm-up*		
1. Squat Circuit	3 $\times$ 1	30
2a. Hang Clean	4 $\times$ 4	60
2b. Tuck Jump	4 $\times$ 3	90
3. Trap Bar Deadlift	3 $\times$ 5	120
4. Bench Pull	5 $\times$ 3	180
5a. Military Press	3 $\times$ 5	30
5b. Chin up	3 $\times$ 8	30
5c. Dumbbell Half get up	3 $\times$ 10	30
Stretch		

*Warm-up details are available in Table 2.

power profiles were retested (MID) and training loads adjusted to ensure subjects continued to train at the appropriate load for their intervention group. They then trained for a further 3 weeks and then in the 10th week final testing

TABLE 2. The warm-up protocol used before testing.

Exercise	Duration
General warm-up	
Jog	5 min
Skip	2 $\times$ 20 m
Sidestep	2 $\times$ 20 m
Carioca	2 $\times$ 20 m
High Knee	2 $\times$ 20 m
Heel Flick	2 $\times$ 20 m
Sumo Squat	1 $\times$ 20 m
Lunge Complex	1 $\times$ 20 m
Inchworm	5 reps
Specific Bench Pull warm-up	
Bar (20 kg)	10 reps
40 kg	5 reps
50 kg	3 reps

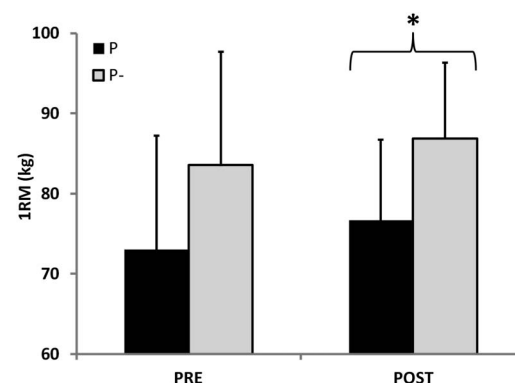


Figure 2. PRE and POST bench pull 1RM performance (in kilograms) for each group (* significant difference from PRE to POST for the cohort as a whole, $p = 0.005$). 1RM, 1 repetition maximum.

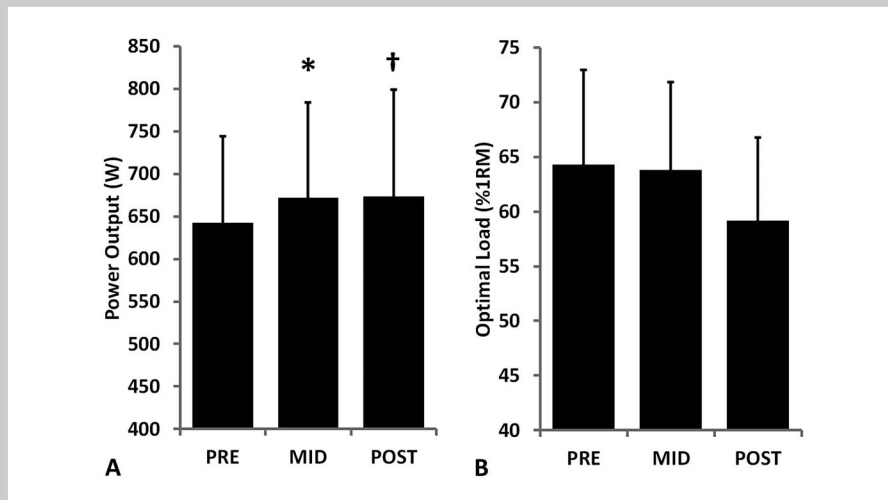


Figure 3. Results of power profile testing for the cohort as a whole: (A) mean peak power output (* significantly different to PRE, $p = 0.002$; † significantly different to PRE, $p = 0.015$); (B) mean load at which peak power was expressed (optimal load). Note that the optimal load is expressed as a percentage of the 1RM at PRE for PRE and MID, and the 1RM at POST for POST. 1RM, 1 repetition maximum.

took place (POST; 1RM test followed by a day's rest and then power profiling).

For each testing session, the subjects performed a standardized full-body general warm-up followed by a specific bar warm-up (Table 2). They then performed the BP testing protocol for the day. Each lift started with the bar stationary beneath the bench with the bar supported at a height such that the subject's arms were straight. The subject then pulled the bar until it contacted the bench and lowered it under control while maintaining contact of their chest, hips, and feet with the bench. For the 1RM testing, the weight was increased in increments of 10 kg then 2.5 kg and subjects were permitted a maximum of 3 attempts at each load with 2–4 minutes of rest between attempts (1). For the power profiling, subjects performed 2 repetitions at each of 30, 40, 50, 60, 70, and 80% of their 1RM (in the ascending order) with 2–4 minutes of rest between sets. The TENDO unit was used to determine the maximum power output for each

load and then the load that permitted the greatest power output (the peak power) was taken to be the optimal load.

The training intervention involved the subjects performing 5 sets of 3 reps in the BP (using the appropriate load for their intervention group) with 3 minutes of rest between sets on 2 separate occasions each week. The frequency and volume of the BP training programme were based on the resistance guidelines of Baechle et al. (2). When performing the BP, training subjects were continually supervised and instructed to lift the bar as fast as possible and to hit the pad on the bench apparatus as hard as possible to ensure there was no unnecessary deceleration of

the bar. The BP training was incorporated within a full-body strength and conditioning programme (consisting of 2 sessions each week), which used lower-body strength and power exercises, upper-body pressing, and trunk stability training. The content of each subject's programme was individualized based on their sporting needs; however, upper-body pulling exercises were standardized between the subjects. An example training programme is depicted in Table 3.

Statistical Analyses

Statistical procedures were performed using SPSS (Version 21; IBM Corp., Armonk, NY). Repeated-measures analysis of variance was used to test for differences in power outputs at each load, peak power output, optimal load, and 1RM across the course of the study (the assumption of sphericity was tested with Mauchly's test of sphericity, and the Greenhouse Geisser adjustment was used if the assumption of sphericity was violated—this was the only case for power output at 50% of 1RM). Post hoc pairwise comparisons were adjusted with Bonferroni corrections to account for multiple comparisons. PRE to POST effect sizes were calculated using Cohen's d following the recommendations and interpretations of Rhea (33). Independent samples T-tests were used to test for differences between the 2 groups at PRE. Significance was determined a priori ($\alpha = 0.05$).

RESULTS

Subjects experienced a modest improvement in their 1RM BP across the course of the study (ie, a significant main effect of time; $F(1,19) = 9.806$, $p = 0.005$, $d = 0.270$; Figure 2). There was a significant effect of time on peak power output for the cohort as a whole ($F(2,38) = 8.991$, $p = 0.001$). Post

TABLE 4. Frequency of the loads at which peak power was produced.*

	30%	40%	50%	60%	70%	80%
PRE	0	0	3	8	8	2
MID	0	0	2	11	6	2
POST	0	0	5	9	6	1

*All values are given as a percentage of the PRE 1 repetition maximum.

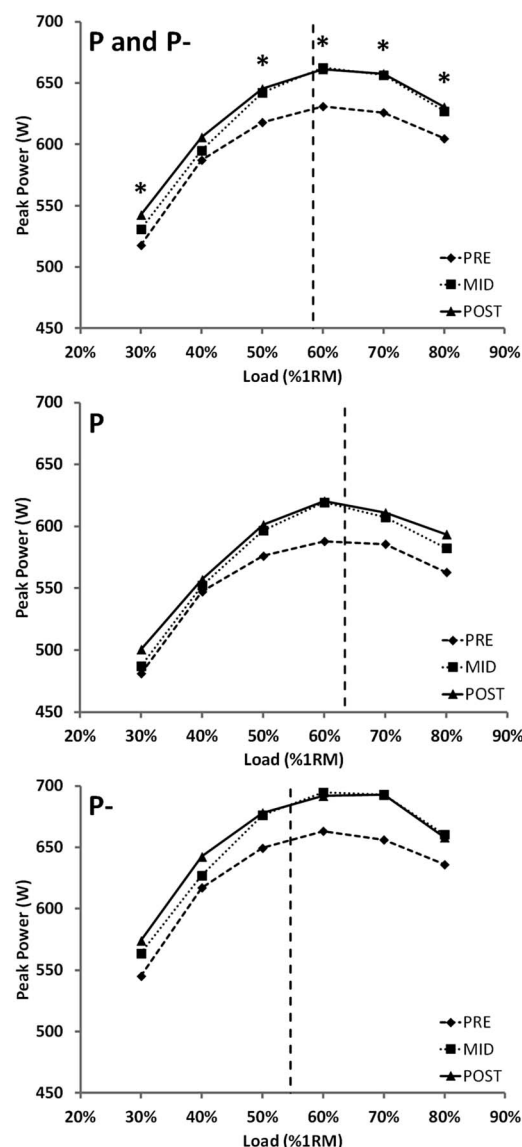


Figure 4. Power profiles for each group (* significant difference from PRE to POST for the cohort as a whole, $p \leq 0.05$). The vertical dashed line represents the mean load that was used for power training during the course of the study for each group.

hoc testing suggested that the cohort significantly increased their peak power output from PRE to MID ($p = 0.002$, $d = 0.290$), but there was no further increase by POST (Figure 3A). The cohort as a whole also experienced a decrease in the optimal load (as a percentage of 1RM) across the course of the study which approached significance (ie, a main effect of time; $F(2,38) = 3.188$, $p = 0.052$, $d = 0.590$; Figure 3B). This decrease was a result of both the increased 1RM (Figure 2) and a decrease in the absolute load at which peak power was expressed (which fell from 64.2% of PRE 1RM to

61.4% of PRE 1RM at POST; see also the frequency of peak power loads in Table 4).

Subjects significantly increased their power outputs at all loads apart from 40% of 1RM (30%: $F(2,38) = 4.889$, $p = 0.013$; 40%: $F(2,38) = 2.436$, $p = 0.101$; 50%: $F(1.497,28.449) = 5.269$, $p = 0.018$; 60%: $F(2,38) = 10.819$, $p < 0.001$; 70%: $F(2,38) = 6.484$, $p = 0.004$; 80%: $F(2,38) = 3.703$, $p = 0.034$; Figure 4). There were no differences for the main effect of group or the interaction effect for any of the variables considered.

DISCUSSION

The subjects who participated in this study (when considered as a whole) experienced a small significant ($p = 0.002$) increase in a peak power output of 4.6% ($d = 0.290$) after 4 weeks of training (which is representative of an upward shift of the complete power-load curve). These results are in broad agreement with other studies that have shown an increase in power output after specific power training (25,26,29). However, a further 3 weeks of the same training failed to provide further increases in power output, despite the fact that the training load was modified to ensure that subjects were training at the desired load. This observation is consistent with other previous power training studies that have demonstrated diminishing returns in terms of improvements in power from 5 weeks onward (10,11).

Cormie et al. (11) found significant ($p \leq 0.05$) increases in the peak power produced in the vertical jump after 5 weeks of training but no further increase at 10 weeks in a group of stronger subjects (able to squat $1.97 \times$ body weight; the weaker subjects experienced a modest further increase at 10 weeks). It was suggested that one explanation for this was the decrease in the absolute strength of the stronger subjects across the course of their study. In contrast, in this study, the subjects experienced a modest increase in their absolute strength in the BP but this did not increase their power output beyond 4 weeks of training.

This study therefore suggests that a significant increase in power output in the BP can be obtained in a reasonably short amount of time by training at, or just below, the optimal load (although the effect size of this increase was small). However, these gains will be exhausted in a short amount of time. This study then further emphasizes the importance of periodization and progressive overload to prevent accommodation and stagnation (37–39) and supports Baker's (4) suggestion that training at the optimal load be reserved for the last few weeks of a training cycle. One idea that was untested within this study, and that should be the focus of future research, is the effect of varying the percentage of 1RM used for power training and whether this is a strategy that can prolong the period of power output improvements leading to greater overall increases in power output.

The secondary aim of this study was to evaluate if there was any difference between training at the optimal load (P), or just below it (P-). Our results suggest that there was no

difference between P and P-. One reason for this could be the fact there are only small differences in power output when using loads close to the optimal load. For instance, Baker and Newton (8) showed how an athlete who trained with 84% of the optimal load could produce 96% of their peak power output. Equally, Baker et al. (7) reported that there was a similarity in the power produced in the range of loads (48–63% of 1RM) surrounding the optimal load (55–59% of 1RM) in the jump squat and Newton et al. (30) found that the greatest mean power was produced at 30 and 45% of 1RM in the bench press throw. These results support the idea that there is a plateau in power outputs when using loads around the optimal load (8, a plateau that was also seen in our results). This in turn suggests that both groups would exhibit similar power outputs relative to their potential maximums during training, thus making the training stimulus markedly similar.

A further interesting result in this study was the decrease in optimal load (as a percentage of 1RM) across the course of the study (a decrease that approached significance; $p = 0.052$). In addition, the effect size of this decrease was largest seen in this study ($d = 0.590$). This decrease was not simply the result of the increased 1RM of the subjects as the absolute value of the optimal load also decreased during the study (Table 4). In contrast to the peak power outputs that did not exhibit a change in the latter half of the study, the optimal load fell throughout the course of the study. This provides tentative support for the notion that the power profile did not remain static in the final 3 weeks of the study but rather experienced a left shift. It can then be speculatively suggested that training around the optimal load may result in both an upward and leftward shift of the power-load curve. There is some disagreement within the literature as to whether the optimal load of stronger athletes is at a higher or lower percentage of 1RM than that of weaker athletes (4,12,24). These results allow us to speculatively suggest that the ability to produce peak power outputs at a relatively lower load is a specific adaptation to training (and in particular to training at or just below the optimal load).

The optimal load for the subjects in this study ranged between 50 and 80% of 1RM with a mean of $64.3\% \pm 9.7$ and $59.1\% \pm 7.6$ of 1RM at PRE and POST, respectively. To our knowledge, there is no research in the BP to support this finding but similar values for optimal load between 50 and 70% of 1RM have been found in other upper-body exercises (6,13). This information will be of value to strength and conditioning coaches particularly if they do not have the required equipment to perform an individualized assessment of their athletes' power profile in the BP.

In this study, we endeavored to recognize the limitations of previous research in this area (14). In particular, there have been weaknesses in the load selection in previous training intervention studies with a number of studies not assessing the actual optimal load of the subjects involved for the movement studied (14). A strength of this study is therefore

that the training load used by each subject was individualized based on their personal power profile. In addition, it is also possible for the optimal load to vary over the length of the study (25); thus, we adjusted the training loads used by the subjects in this study based on the MID results to ensure that subjects continued to train at the desired load. A potential limitation of this study is the uneven group numbers because of the exclusion of 4 subjects from the study for not meeting the training threshold requirements (3 of the excluded subjects being from the P group). However, a sensitivity analysis showed that if the group numbers were balanced by including the 3 subjects excluded from the P group and if their peak power increased by 20% (the maximum increase seen by any subject in the study), there would still have been no significant differences in the increase in peak power between P and P-.

PRACTICAL APPLICATIONS

The results of this study suggest that small but significant improvements in power output can be realized in only a few weeks of specific power training with the optimal load, and that there is no benefit in prolonging this type of training. Thus, the results of this study tend to support Baker's (4) recommendation that power training with the optimal load only be performed in the last few weeks of a training cycle. In addition, these results also support the idea that similar improvements in power output can be realized by training with loads that are below the optimal load (8). A further finding of this study was a relative decrease in optimal load across the course of the study. Strength and conditioning coaches should be cognizant of the fact that power training with the optimal load may also cause a left shift of the power-load curve toward peak power being expressed at relatively lighter loads and ensure that this is a desired training effect. Finally, this study also supports previous studies that suggest that the optimal load for upper-body exercises is in the range of 50–70% of 1RM. This information will be of use to strength and conditioning coaches and especially to those who do not have the facility to evaluate the individual power profiles of their athletes.

ACKNOWLEDGMENTS

There are no conflicts of interest. The results of this study do not constitute endorsement of any product by the authors or the National Strength and Conditioning Association.

REFERENCES

1. Appleby, B, Newton, RU, and Cormie, P. Changes in strength over a 2-year period in professional rugby union players. *J Strength Cond Res* 26: 2538–2546, 2012.
2. Baechle, TR, Earle, RW, and Wathen, D. Resistance training. In: *Essentials of Strength Training and Conditioning* (2nd ed.). Champaign, IL: Human Kinetics, 2000. pp. 395–426.
3. Baker, D. Comparison of upper-body strength and power between professional and college-aged rugby league players. *J Strength Cond Res* 15: 30–35, 2001.

4. Baker, D. A series of studies on the training of high-intensity muscle power in rugby league football players. *J Strength Cond Res* 15: 198–209, 2001.
5. Baker, D. Differences in strength and power among junior-high, senior-high, college-aged, and elite professional rugby league players. *J Strength Cond Res* 16: 581–585, 2002.
6. Baker, D, Nance, S, and Moore, M. The load that maximizes the average mechanical power output during explosive bench press throws in highly trained athletes. *J Strength Cond Res* 15: 20–24, 2001.
7. Baker, D, Nance, S, and Moore, M. The load that maximizes the average mechanical power output during jump squats in power-trained athletes. *J Strength Cond Res* 15: 92–97, 2001.
8. Baker, D and Newton, RU. Methods to increase the Effectiveness of maximal power training for the upper body. *Strength Cond J* 27: 24–32, 2005.
9. Clark, RA, Bryant, AL, and Humphries, B. A comparison of force curve profiles between the bench press and ballistic bench throws. *J Strength Cond Res* 22: 1755–1759, 2008.
10. Cormie, P, McGuigan, MR, and Newton, RU. Adaptations in athletic performance after ballistic power versus strength training. *Med Sci Sports Exerc* 42: 1582–1598, 2010.
11. Cormie, P, McGuigan, MR, and Newton, RU. Influence of strength on magnitude and mechanisms of adaptation to power training. *Med Sci Sports Exerc* 42: 1566–1581, 2010.
12. Cormie, P, McGuigan, MR, and Newton, RU. Developing maximal neuromuscular power: Part 2-training considerations for improving maximal power production. *Sports Med* 41: 125–146, 2011.
13. Cronin, J, McNair, PJ, and Marshall, RN. Velocity specificity, combination training and sport specific tasks. *J Sci Med Sport* 4: 168–178, 2001.
14. Cronin, J and Sleivert, G. Challenges in understanding the influence of maximal power training on improving athletic performance. *Sports Med* 35: 213–234, 2005.
15. García-Pallarés, J, Sánchez-Medina, L, Carrasco, L, Díaz, A, and Izquierdo, M. Endurance and neuromuscular changes in world-class level kayakers during a periodized training cycle. *Eur J Appl Physiol* 106: 629–638, 2009.
16. Garthe, I, Raastad, T, and Sundgot-Borgen, J. Long-term effect of weight loss on body composition and performance in elite athletes. *Int J Sport Nutr Exerc Metab* 21: 426, 2011.
17. Hansen, KT, Cronin, JB, Pickering, SL, and Douglas, L. Do force-time and power-time measures in a loaded jump squat differentiate between speed performance and playing level in elite and elite junior rugby union players? *J Strength Cond Res* 25: 2382–2391, 2011.
18. Hedeker, D, Gibbons, RD, and Waternaux, C. Sample size estimation for longitudinal designs with attrition: Comparing time-related contrasts between two groups. *J Educ Behav Stat* 24: 70–93, 1999.
19. Hermassi, S, Chelly, MS, Fathloun, M, and Shephard, RJ. The effect of heavy-vs. moderate-load training on the development of strength, power, and throwing ball velocity in male handball players. *J Strength Cond Res* 24: 2408–2418, 2010.
20. Hermassi, S, Chelly, MS, Tabka, Z, Shephard, RJ, and Chamari, K. Effects of 8-week in-season upper and lower limb heavy resistance training on the peak power, throwing velocity, and sprint performance of elite male handball players. *J Strength Cond Res* 25: 2424–2433, 2011.
21. Ivey, P, Oakley, J, and Hagerman, P. Strength training for the preparatory phase in collegiate women's rowing. *Strength Cond J* 26: 10–15, 2004.
22. Jennings, CL, Viljoen, W, Durandt, J, and Lambert, MI. The reliability of the FittroDyne as a measure of muscle power. *J Strength Cond Res* 19: 859–863, 2005.
23. Kaneko, M, Fuchimoto, T, Toji, H, and Suei, K. Training effect of different loads on the force-velocity relationship and mechanical power output in human muscle. *Scand J Sports Sci* 5: 50–55, 1983.
24. Kawamori, N and Haff, GG. The optimal training load for the development of muscular power. *J Strength Cond Res* 18: 675–684, 2004.
25. Mayhew, JL, Ware, JS, Johns, RA, and Bemben, MG. Changes in upper body power following heavy-resistance strength training in college men. *Int J Sports Med* 18: 516–520, 1997.
26. McBride, JM, Triplett-McBride, T, Davie, A, and Newton, RU. The effect of heavy- vs. light-load jump squats on the development of strength, power, and speed. *J Strength Cond Res* 16: 75–82, 2002.
27. McNeely, E, Sandler, D, and Bamel, S. Strength and power goals for Competitive Rowers. *Strength Cond J* 27: 10–15, 2005.
28. Naclerio, F, Faigenbaum, AD, Larumbe-Zabala, E, Perez-Bibao, T, Kang, J, Ratamess, NA, and Triplett, NT. Effects of different resistance training volumes on strength and power in team sport athletes. *J Strength Cond Res* 27: 1832–1840, 2013.
29. Newton, RU, Kraemer, WJ, and Häkkinen, K. Effects of ballistic training on preseason preparation of elite volleyball players. *Med Sci Sports Exerc* 31: 323–330, 1999.
30. Newton, RU, Murphy, AJ, Humphries, BJ, Wilson, GJ, Kraemer, WJ, and Häkkinen, K. Influence of load and stretch shortening cycle on the kinematics, kinetics and muscle activation that occurs during explosive upper-body movements. *Eur J Appl Physiol* 75: 333–342, 1997.
31. Overall, JE and Doyle, SR. Estimating sample sizes for repeated measurement designs. *Control Clin Trials* 15: 100–123, 1994.
32. Requena, B, García, I, Requena, F, de Villarreal, ES, and Cronin, JB. Relationship between traditional and ballistic squat exercise with vertical jumping and maximal sprinting. *J Strength Cond Res* 25: 2193–2204, 2011.
33. Rhea, MR. Determining the magnitude of treatment effects in strength training research through the use of the effect size. *J Strength Cond Res* 18: 918–920, 2004.
34. Sleivert, G and Taingahue, M. The relationship between maximal jump-squat power and sprint acceleration in athletes. *Eur J Appl Physiol* 91: 46–52, 2004.
35. Smilios, I, Sotiropoulos, K, Christou, M, Douda, H, Spaias, A, and Tokmakidis, SP. Maximum power training load determination and its effects on load-power relationship, maximum strength, and vertical jump performance. *J Strength Cond Res* 27: 1223–1233, 2013.
36. Stock, MS, Beck, TW, DeFreitas, JM, and Dillon, MA. Test-retest reliability of barbell velocity during the free-weight bench-press exercise. *J Strength Cond Res* 25: 171–177, 2011.
37. Stone, MH, Keith, RE, Kearney, JT, Fleck, SJ, Wilson, GD, and Triplett, NT. Overtraining: A review of the signs, symptoms and possible causes. *J Strength Cond Res* 5: 35–50, 1991.
38. Stone, MH, O'Bryant, H, and Garhammer, J. A hypothetical model for strength training. *J Sports Med Phys Fitness* 21: 342, 1981.
39. Stowers, T, McMillan, J, Scala, D, Davis, V, Wilson, D, and Stone, M. The short-term effects of three different strength-power training methods. *Strength Cond J* 5: 24–27, 1983.
40. Wilson, GJ, Newton, RU, Murphy, AJ, and Humphries, BJ. The optimal training load for the development of dynamic athletic performance. *Med Sci Sports Exerc* 25: 1279–1286, 1993.